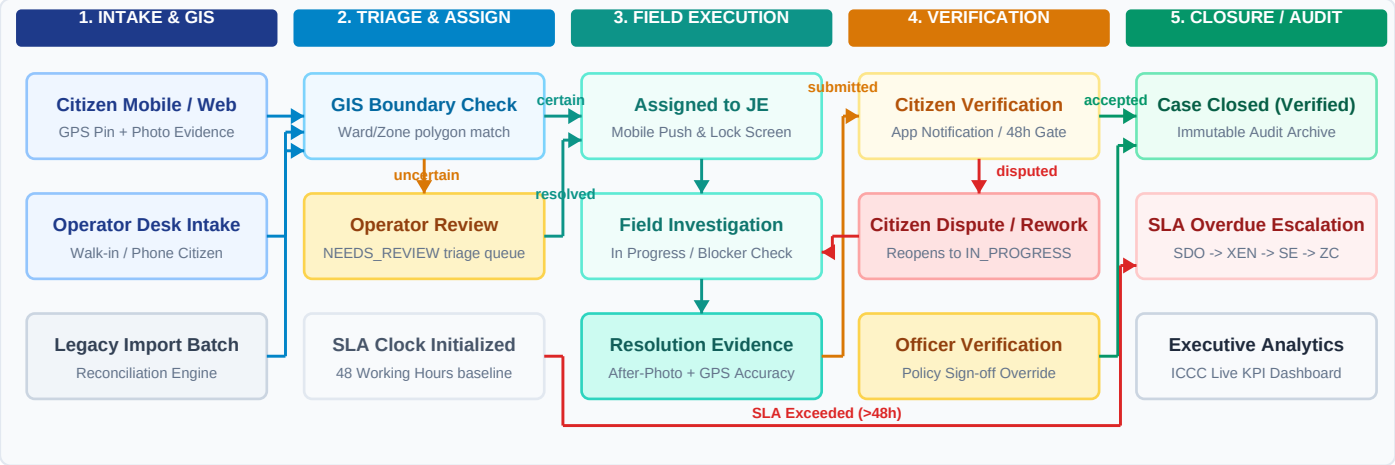
MUNICIPAL CORPORATION LUDHIANA (MCL)

Integrated Command & Control Centre (ICCC) — Smart Complaint Management System

Comprehensive System Architecture, Operational Workflows, Data Movement & State Machine Specification

1. Executive Operational Workflow & System Topology

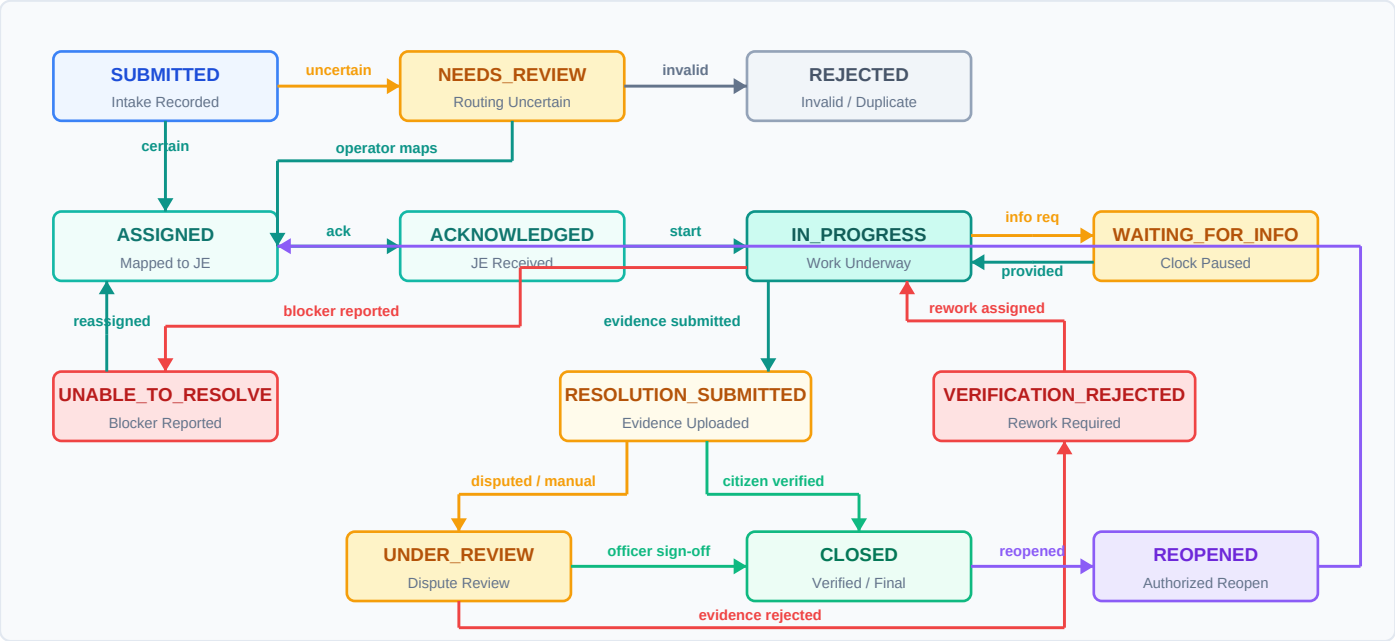
The MCL ICCC Complaint Management System automates the civic issue resolution lifecycle across Municipal Corporation Ludhiana. The architecture unifies citizen grievance intake, spatial geocoded boundary routing, mobile field dispatch to Junior Engineers (JEs), strict verification gates (preventing premature ticket closure), and automated 48-working-hour SLA escalation monitoring into an auditable, single-source-of-truth platform.



Phase	Operational Actor & Target	Core Processing & Governance Rules	Data Movement & System State
1. Intake & Ingestion	Citizens (Web/Mobile), Desk Operators, Legacy Import	Account-based registration. Captures category (15 standard types), photo/video, and map pin. Reverse geocoded against MCL GIS layer.	Payload persists to complaints; generates unique public complaint #; state → SUBMITTED.
2. Spatial Triage	GIS Ingestion Engine / Operator Queue	Evaluates spatial polygon and JE roster mappings. High-certainty mappings auto-assign. Ambiguous cases route to Operator Queue.	High confidence → ASSIGNED; Uncertain boundary/roster → NEEDS_REVIEW.
3. Field Execution	Junior Engineer (JE) / Field Resolver	Instant mobile lock-screen push. JE acknowledges assignment, conducts on-site remediation, or reports blockers.	Transitions: ACKNOWLEDGED → IN_PROGRESS. Blockers transition to UNABLE_TO_RESOLVE.
4. Resolution & Evidence	Junior Engineer (JE)	Mandatory resolution report: after-photos uploaded to private storage + resolver GPS coordinates with device accuracy.	Transitions to RESOLUTION_SUBMITTED. **Governance Rule:** Resolver cannot self-close complaint.
5. Verification Gate	Citizen / Authorized Supervisory Officer	Citizen receives verification push. Citizen can accept (closes ticket) or reject (triggers mandatory rework loop).	Accepted → CLOSED; Disputed → UNDER_REVIEW / VERIFICATION_REJECTED → Rework.
6. SLA & Escalation	ICCC SLA Engine & Officer Hierarchy	48 working-hour countdown using MCL calendar (excluding nights/holidays). Automated alerts escalate through hierarchy.	Hierarchy: JE → SDO → XEN → SE → ZC → JC. Case marked Overdue; never auto-closed.

2. Authoritative 13-State Machine & Transition Architecture

The complaint lifecycle is governed by a strict, backend-enforced deterministic finite state machine (FSM). Arbitrary client status updates are prohibited; every transition requires role authorization, required input validations, and produces an immutable record in complaint_status_history.

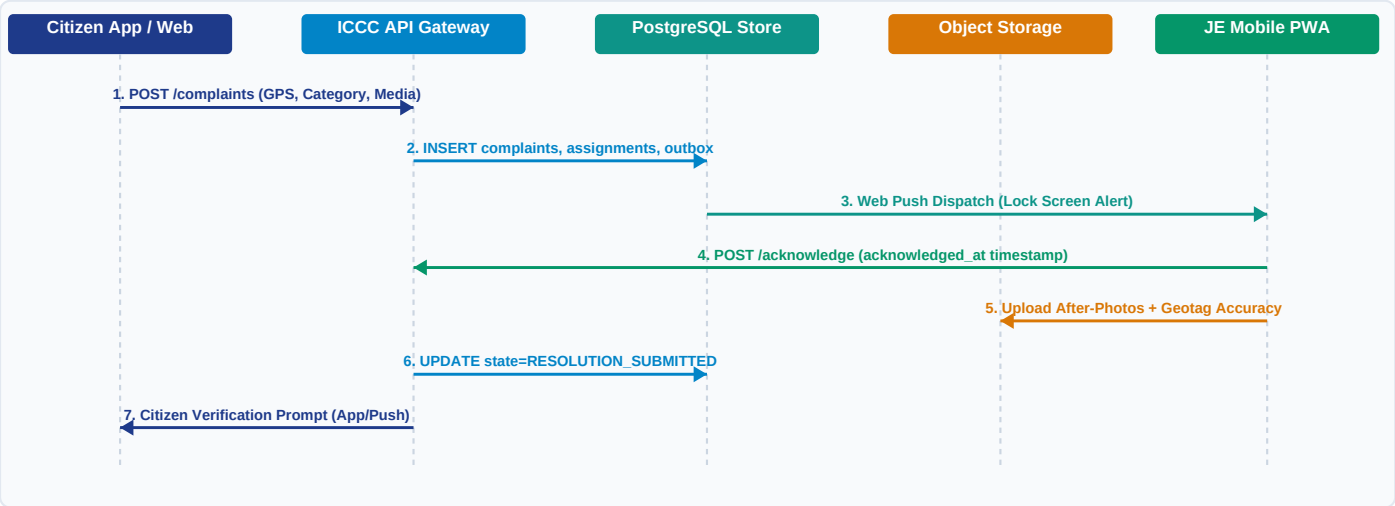


System State	Operational Meaning	Valid Inbound From	Valid Outbound To	Authorized Actor & Transition Guards
SUBMITTED	Grievance received and assigned unique tracking #.	*Initial Intake*	NEEDS_REVIEW, ASSIGNED, REJECTED	System / Ingestion Gateway. Validates category, description, and initial media.
NEEDS_REVIEW	Spatial boundary or JE roster mapping uncertain.	SUBMITTED	ASSIGNED, REJECTED	Desk Operator. Requires manual ward/branch assignment justification.
ASSIGNED	Mapped to responsible JE in authoritative roster.	SUBMITTED, NEEDS_REVIEW, UNABLE_TO_RESOLVE, REOPENED	ACKNOWLEDGED, NEEDS_REVIEW (Reassign)	System / Operator. Dispatches mobile push and starts 48h SLA clock.
ACKNOWLEDGED	JE confirmed receipt of field assignment.	ASSIGNED	IN_PROGRESS, UNABLE_TO_RESOLVE	Assigned JE. Records acknowledged_at timestamp for field latency tracking.
IN_PROGRESS	Field investigation or remedial works active.	ACKNOWLEDGED, WAITING_FOR_INFO, VERIFICATION_REJECTED	WAITING_FOR_INFO, UNABLE_TO_RESOLVE, RESOLUTION_SUBMITTED	Assigned JE. Can upload interim progress notes and inspection media.
WAITING_FOR_INFO	Awaiting additional detail or access from citizen.	IN_PROGRESS	IN_PROGRESS	Assigned JE. **SLA Pause Rule:** Pauses working-hour clock until response.
UNABLE_TO_RESOLVE	Statutory, technical, or legal blocker encountered.	IN_PROGRESS, ACKNOWLEDGED	ASSIGNED (Reassigned), UNDER_REVIEW	Assigned JE. Mandatory blocker rationale; alerts SDO supervisor.
RESOLUTION_SUBMITTED	Remediation complete; evidence package attached.	IN_PROGRESS	CLOSED, UNDER_REVIEW	Assigned JE. Requires after-photos + device GPS accuracy metadata. **Does not close case.**
UNDER_REVIEW	Disputed resolution or manual officer review active.	RESOLUTION_SUBMITTED, UNABLE_TO_RESOLVE	CLOSED, VERIFICATION_REJECTED	Desk Operator / SDO Officer. Reviewing evidence validity or citizen challenge.
VERIFICATION_REJECTED	Resolution evidence rejected or citizen disputed.	UNDER_REVIEW	IN_PROGRESS	Citizen / Supervisory Officer. Reverts case to JE with mandatory rework memo.
CLOSED	Grievance resolved and verified by citizen or officer.	RESOLUTION_SUBMITTED, UNDER_REVIEW	REOPENED	Citizen (App verification) or Authorized Officer. Stops active SLA reminders.
REOPENED	Resolved issue recurred; case resumed.	CLOSED	ASSIGNED	Authorized Officer / Citizen within allowable challenge window. Preserves prior history.
REJECTED	Out-of-jurisdiction, duplicate, or invalid issue.	SUBMITTED, NEEDS_REVIEW	*Terminal*	Desk Operator / Supervisor. Mandatory written rejection reason & citizen alert.

Core Governance Directives (from Mplan.md):
- ***Resolution ≠ Closure:*** *A JE claiming resolution only moves status to RESOLUTION_SUBMITTED. Closure requires citizen verification or authorized officer certification.*
- ***No Auto-Closure on Inactivity:*** *If a citizen does not verify within the reminder period, complaints transfer to an Operator/Officer review queue; they are never silently auto-closed.*
- ***Immutable Audit Trail:*** *Every status transition records actor_user_id, previous_status, new_status, reason, and server timestamp. History cannot be overwritten.*

3. Data Movement & Subsystem Interaction Architecture

Data movement across the MCL ICCC platform operates through decoupled, asynchronous event channels. The diagram and specifications below detail the exact payload flows, database transactions, object storage exchanges, and push notification events across the system boundary.



A. Intake Ingestion & Spatial Geocoding Pipeline

1. **Payload Submission:** Citizen client posts multipart JSON containing category code, description, reverse-geocoded address string, map pin coordinates (latitude, longitude), and initial photo media.
2. **Spatial Boundary Verification:** API Gateway executes a spatial containment query against the MCL boundary shapefiles to identify the target Zone and Ward. Coordinates falling outside Ludhiana municipality boundaries are flagged.
3. **Roster Matching:** The system queries staff_area_assignments matching (branch_id, zone_id, ward_id). If exactly one active JE is found with high confidence, assigned_je_id is populated. If unmapped or conflicting, the record is flagged for operator triage.
4. **Atomic Write:** Database inserts the complaint record, creates an initial row in complaint_assignments, logs state in complaint_status_history, and queues a notification_events row in a single ACID transaction.

B. Mobile Dispatch & Lock-Screen Notification Pipeline

1. **Outbox Processing:** An asynchronous notification worker reads pending notification_events and looks up the assigned JE's active endpoints in push_subscriptions.
2. **High-Urgency Web Push:** Dispatches RFC-8291/RFC-8292 Web Push payloads with urgency: 'high' and topic: 'mcl-civic-alert'. Mobile service workers trigger persistent notifications with interactive action buttons ('Acknowledge', 'View Details').
3. **Receipt Handshake:** Upon opening, the field PWA issues POST /api/complaints/:id/acknowledge, updating acknowledged_at = NOW() and transitioning state to ACKNOWLEDGED.

C. Field Resolution Evidence & Geotag Integrity Flow

1. **Evidence Packaging:** When work is completed, the JE captures after-photos via the mobile browser/PWA camera interface. The client queries the HTML5 Geolocation API with enableHighAccuracy: true.
2. **Secure Media Ingestion:** Image buffers stream directly to private object storage under segregated tenant prefixes /evidence/{complaint_id}/{uuid}.jpg.
3. **Metadata Recording:** complaint_media stores the private storage key, uploader UID, device_latitude, device_longitude, and gps_accuracy_meters. The server logs whether GPS was captured, denied, or degraded.

D. Working-Hour SLA Countdown & Escalation Engine

1. **SLA Baseline:** 48 working hours, calculated against MCL operational hours (09:00 to 17:00 IST, Monday through Friday, excluding official Punjab Government Gazetted holidays).
2. **Clock Pause Rules:** Moving to WAITING_FOR_INFORMATION formally halts the SLA clock until citizen feedback is submitted.
3. **Hierarchical Escalation:** A 5-minute background cron evaluates open complaints against due_at. When elapsed working time crosses thresholds, automated push/in-app escalation alerts fire across the administrative chain: **JE (+0h)** → **SDO (+12h overdue)** → **XEN (+24h overdue)** → **SE (+48h overdue)** → **Zonal Commissioner** → **Joint Commissioner**.

E. Operator Shift Handover & Audit Integrity

1. **Individual Attribution:** Operator accounts are strictly individual (no shared desk logins). Every queue interaction records operator_user_id.
2. **Shift Handover Record:** At shift conclusion, outgoing operators generate a formal handover ledger (shift_handover_records) documenting total pending cases, critical overdue items, follow-up callbacks scheduled, and shift transition notes. The incoming operator cryptographically acknowledges receipt upon login.

4. Role-Based Access Control (RBAC) & Actor Matrix

Access control is enforced at both API route middleware and PostgreSQL database policy layers. User roles define actionable permissions, while geographical and branch assignments restrict data query visibility.

System Role	Jurisdictional Scope	Permitted State Actions	Data Visibility & Governance Limits
Citizen	Personal records only	SUBMIT, WAITING_FOR_INFO (Reply), VERIFY (Accept/Dispute)	Restricted strictly to personal complaints. Cannot view officer internal notes or other citizens' data.
Junior Engineer (JE)	Assigned Wards / Branch	ACKNOWLEDGE, IN_PROGRESS, WAITING_FOR_INFO, UNABLE_TO_RESOLVE, RESOLUTION_SUBMITTED	Restricted to complaints assigned within designated ward roster. Cannot self-close or delete complaints.
Desk Operator	All Zones / MCL-wide	SUBMIT (Citizen on behalf), REASSIGN, FOLLOW_UP, REVIEW_TRIAGE, REJECT	Cannot bypass state machine or perform silent edits. All routing corrections log original and updated values.
Sub-Divisional Officer (SDO)	Sub-Division / Zone	REASSIGN, RESOLVE_BLOCKER, OVERRIDE_VERIFICATION, ESCALATE	Supervises assigned JEs; reviews disputed resolution evidence and high-aging bottlenecks.
Executive Engineer (XEN)	Division / Branch-wide	ESCALATION_REVIEW, INTERVENE, REALLOCATE_RESOURCE	Branch-level oversight; receives SLA breach escalations; audits repeat complaint hotspots.
Superintending Engineer (SE)	Branch Circle	CIRCLE_OVERSIGHT, INTER_BRANCH_COORDINATION	Circle-level performance monitoring; reviews chronic infrastructure failures across divisions.
Zonal / Joint Comm. (ZC/JC)	Zonal / City-wide Executive	CITY_AUDIT, POLICY_OVERRIDE, FINAL_DISPOSITION	Full executive visibility; live Command Wall dashboards; policy sign-off on mass exceptions.
System Administrator	System-wide Infrastructure	USER_MGMT, ROSTER_CONFIG, CALENDAR_MGMT, DATA_IMPORT	Restricted from mutating operational complaint lifecycle directly; actions strictly audit-logged.

5. Technical Relational Data Schemas & Entity Architecture

The relational schema is implemented in PostgreSQL, utilizing UUID primary keys, foreign key constraints, and append-only history ledgers to guarantee strict relational integrity and auditability.

Table Name	Primary & Foreign Keys	Core Relational Attributes	Storage & Integrity Purpose
complaints	PK: id (UUID) FK: citizen_id, category_id, ward_id, assigned_je_id	complaint_number (UNIQUE), description, current_status, latitude, longitude, submitted_at, due_at, resolved_at, closed_at	Central operational ledger. Single source of truth for grievance state and SLA calculations.
complaint_status_history	PK: id (UUID) FK: complaint_id, actor_user_id	previous_status, new_status, action_reason, created_at	Append-only audit ledger tracking every state transition with actor attribution.
complaint_assignments	PK: id (UUID) FK: complaint_id, assigned_to_user_id, assigned_by_user_id	assignment_reason, assigned_at, acknowledged_at, is_current	Historical and active officer assignment log. Captures field acknowledgment timestamps.
complaint_media	PK: id (UUID) FK: complaint_id, uploaded_by_user_id	storage_key, media_type, purpose (intake/progress/resolution), device_lat, device_lng, accuracy_meters	Metadata index for evidence assets stored in private object storage.
complaint_verifications	PK: id (UUID) FK: complaint_id, verifier_user_id	verifier_role (citizen/officer), result (accepted/rejected), dispute_comments, verified_at	Formal audit sign-off record required before tickets transition to CLOSED.
shift_handover_records	PK: id (UUID) FK: outgoing_operator_id, incoming_operator_id	shift_started_at, shift_ended_at, pending_cases_count, overdue_cases_count, handover_notes	Operator accountability ledger guaranteeing shift continuity without shared logins.
notification_events	PK: id (UUID) FK: complaint_id, recipient_user_id	idempotency_key (UNIQUE), channel (in_app/push), event_type, delivery_state, attempt_count	Transactional outbox decoupling database commits from third-party Web Push providers.
audit_logs	PK: id (UUID) FK: actor_user_id	action_type, entity_type, entity_id, diff_json, ip_address, created_at	Immutable security audit log capturing metadata corrections, role changes, and administrative actions.

Document Sign-off & Baseline Confirmation: *This document establishes the operational workflow and architectural contract for the MCL ICCC Complaint Management System. Any modifications to state transitions, SLA thresholds, or verification gates require versioned approval from the Municipal Corporation of Ludhiana Operations Board.*